REGULATORY DOSSIER SUBMISSION
Dossier ID: DOS-07F23D73B0
Country: Uganda
Submission Date: 2025-05-11
Product: Codenova (codeine) | ATC: M01AE01 | Strength: 200 mg
Applicant: United Therapeutics | Manufacturer: Nile Generic Medicines
Policy Label: reject_and_return | Risk Score: 0.92
AMR Stewardship: AWaRe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable | Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information
Labels: presence=present; length=length_ok; correctness=correct
This dossier constitutes a formal application for Marketing Authorization of Codenova (codeine) submitted on 2025-05-11 to the national regulatory authority of Uganda. The applicant, United Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 200 mg, classified under ATC code M01AE01. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. All referenced documents are available in the dossier annex and traceable by section identifier. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. A comprehensive table of contents and glossary of abbreviations are provided to facilitate regulatory review. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Regional administrative forms were checked for signature completeness and dossier version alignment. Risk

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence
Labels: presence=present; length=length_ok; correctness=correct
The primary commercial manufacturing site for the finished pharmaceutical product is Nile Generic Medicines, located in Botswana. The facility was subject to a comprehensive GMP inspection by a recognized stringent regulatory authority. The latest inspection concluded on 2024-10-22, resulting in a formal compliance status of ’compliant’. The active GMP certificate number is GMP-261745, which remains valid until 2026-03-25. Site master files, a summary of recent inspection observations, and evidence of completed Corrective and Preventive Actions (CAPA) are included in the annex. The applicant confirms alignment with applicable technical guidance and regional submission standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system

consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Method validation records and quality controls were reviewed for internal consistency. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoint

[1] m1_product_information - Product Information and Naming
Labels: presence=present; length=length_ok; correctness=correct
The proposed proprietary name for this medicinal product is Codenova, containing the active pharmaceutical ingredient codeine. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. Method validation records and quality controls were reviewed for internal consistency. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. All referenced documents are available in the dossier annex and traceable by section identifier. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Stability testing was conducted in climatic

[2] m2_quality_overall_summary - Quality Overall Summary
Labels: presence=present; length=length_ok; correctness=correct
This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for codeine. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The risk management plan includes routine pharmacovigilance and pr

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary
Labels: presence=present; length=length_ok; correctness=correct
The Clinical Overview synthesizes the efficacy and safety data supporting the use of Codenova for the treatment of community acquired pneumonia. The pivotal clinical development program comprised 2 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is ’endpoint_met’. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. No new safety signals were identified during the extended open-label follow-up period. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. Clinical interpretation focuses on whether efficacy, safety, and benefit-risk conclusions are supported by the submitted evidence. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication.
Stability testing was conducted in climatic zones III and IV, reflecting

[3] m3_api_quality - API Quality and Control Strategy
Labels: presence=present; length=length_ok; correctness=correct
The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Batch analysis, validation, and impurity management records support the proposed API quality framework. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. The API package links specification limits to validated analytical methods and impurity-control decisions. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The applicant confirms alignment with applicable technical guidance and regional submission standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Critical material attributes and control strategy elements were reviewed for consistency across the quality dossier. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles.

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[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls
Labels: presence=present; length=length_ok; correctness=correct
The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Method validation records and quality controls were reviewed for internal consistency. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. All referenced documents are available in the dossier annex and traceable by section identifier. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The applicant confirms alignment with applicable technical guidance and regional submission standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confi

[3] m3_stability - Stability and Shelf-Life Justification
Labels: presence=present; length=length_ok; correctness=correct
A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40℃/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The container closure system consists of PVC/PVDC-Aluminum blisters

[4] m4_nonclinical_summary - Nonclinical Study Summary
Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. The applicant confirms alignment with applicable technical guidance and regional submission standards. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. All referenced documents are available in the dossier annex and traceable by section identifier. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations w

[5] m5_trial_listing - Tabular Listing of Clinical Studies
Labels: presence=present; length=length_ok; correctness=correct
The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (community acquired pneumonia). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Risk mitigation activities are docum

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports
Labels: presence=present; length=length_ok; correctness=correct
Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of ’endpoint_met’, demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS)

aligned with ICH Q10. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) ve

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence
Labels: presence=present; length=length_ok; correctness=correct
Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. All referenced documents are available in the dossier annex and traceable by section identifier. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Risk mitigation activities